Autonomous Campus Mobility Services Using Driverless Taxi

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***Abstract*— In this paper, we present a driverless taxi system for autonomous campus mobility services. College campuses have unique mobility requirements in terms of layout, population, and demand and patterns. It is typically recommended to minimize the presence of private automobiles on campuses due to teaching and research disturbances, visual degradation from parking provision, environmental pollution, and negative health effects. As an alternative to private automobiles, shared mobility systems have been considered for both campus and urban transportation. Conventional shuttle systems suffer from the first and last mile problem. A bicycle and pedestrian friendly policy is not a generalizable solution for all geographic locations and campus layouts. We suggest a driverless taxi service as an alternative point-to-point shared mobility system for campuses. We have demonstrated the feasibility of this service on a 4.5-km campus road at Seoul National University. The service has covered over 10 000 km autonomously since the first public demonstration was made in November 2015.**

***Index Terms*— Autonomous driving vehicle, campus mobility service, driverless taxi, one-way shared urban mobility.**

I. Introduction

THE paradigm of urban mobility has shifted significantly over the past few decades. In particular, private automobiles have been revisited and reconsidered from several perspectives such as traffic congestion, pollution, vehicle under-utilization and parking space inefficiency [1]. Shared mobility has risen as a possible alternative solution for urban mobility problems due to its individual and societal advantages in terms of parking, maintenance cost, and environmental impact [2]. Shared mobility has been provided in two ways: two-way and one-way models. In a two-way model, such as a traditional car rental system, a customer can use a car and pass it on to the next customer. The one-way shared mobility model includes car-pooling and public transportation systems, such as buses and urban trains.

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In the last decade, new types of one-way shared mobility systems have emerged as a compromise between the pros and cons of conventional shared mobility systems. Car2Go (one-way car rental system), OV-Fiets [3] (one-way bicycle rental system for public transit), and Uber and Lyft (one-way ride-sharing and demand-responsive transportation systems) are good examples of successfully implemented systems. Furthermore, smart devices that can access geographic information enable one-way ride-sharing systems to increase car-utilization by matching the supply and demand for dynamic point-to-point transportation in real-time [4]. These smart devices have also been used to improve conventional taxi systems in a similar manner [5]. However, there are still several problems to solve for one-way shared mobility systems, including the first and last mile problem [6] and vehicle asymmetry between supply and demand [7].

The development of autonomous vehicle technology has attracted significant interests for its potential to improve the shared mobility. A lightweight autonomous electric vehicle was proposed as a means of providing first and last mile transportation services [8]. In order to redistribute the imbalanced fleet allocation caused by asymmetric trips, autonomous vehicles have been proposed as a method of automated rebalancing [9]. Driverless buses and urban trains have been proposed and operated to provide safety enhancements and capacity on demand without requiring additional operational staff [10], [11]. In this paper, we extend the role of autonomous vehicle technology to one-way point-to-point shared mobility systems, such as a driverless taxi, particularly for confined areas such as college campuses.

College campuses have unique transportation requirements in terms of layout, population, and spatio-temporal mobility patterns. Commuting by private automobiles must be reduced due to teaching and research disturbances, pollution of the natural environment, visual degradation caused by parking provision, and nagative health effects on staff and students [12], [13]. Similar to other public bus systems, campus shuttle systems encounter the first and last mile problem. Bicycle and pedestrian friendly policies are not a generalizable solution, especially in large, sloped, or rural campuses. Alternatively, a driverless taxi provides point-to-point mobility while increasing car-utilization and reaping the benefits of reducing private automobile commutings

We propose an Autonomous Campus Mobility Service (ACMS) using a driverless taxi. We implemented the

TABLE I
SHARED URBAN MOBILITY SERVICES: FIRST MILE PROBLEM, LAST MILE PROBLEM, REBALANCING, POINT-TO-POINT, AND AUTOMATION

Model	Type	Service	*FM*	*LM*	*RB*	*PP*	*AT*	Scenario
Two-way	Car rental	Zipcar, Smove			✓			Need to return the car to the next user
One-way	Public transportation	Bus			✓			Circulation at bus stops (tens of passengers)
		Urban train			✓			Circulation at stations (hundreds of passengers)
		Taxi	✓	✓	✓	✓		Demand-responsive public transport
	Car rental	Car2go, DriveNow						No need to return the car to the next user
	Mobility-on-demand [7]	OV-Fiets [3]	✓	✓				Mobility btw. transit station and home/workplace
		Autonomy for MoD [14]	✓	✓	✓		✓	MoD service with automated re-balancing
	Dynamic ride-sharing [15]	Uber, Lyft	✓	✓	✓	✓		Dynamic and real-time on-demand ride-sharing
	Smart taxi [5], [16]	Kakao taxi	✓	✓	✓	✓		Real-time taxi assignment based on location
	Driverless train [11]	London DLR, SG MRT			✓		✓	Safety enhancement and capacity on demand
	Driverless bus	CityMobil2 [17]	✓	✓	✓		✓	Collective MOD on public road
	Driverless golf car	SMART FM project [18]			✓		✓	Collective MOD at public park
	Driverless taxi	ACMS (Proposed)	✓	✓	✓	✓	✓	Campus mobility service

service on a 4.5 km campus road at Seoul National University, and the service is currently in operation. The contributions of this paper can be summarized as follows:

- An autonomous campus mobility service using a driverless taxi is proposed and deployed as a real service.
- Test results from the real campus road are reported with passenger satisfaction rating, feedback, and additional findings.

The remainder of this paper is organized as follows. Section II reviews exist literature. Section III describes the autonomous campus mobility service. Section VI provides the details of a driverless vehicle for CMS. Section V presents test results. Section VI contains our conclusion.

II. RELATED WORKS

In this section, we review the literature on automation and autonomous vehicle technologies for shared mobility.

Autonomous vehicles could serve as a new mobility paradigm to offer customized and demand-responsive transport services [19], which could also contribute significantly to shared mobility [14]. In this paper, shared mobility is defined as a transportation service that is shared among users, including public transport, taxis, car-sharing, ride-sharing, and shuttle services. Shared mobility has been proposed as a promising solution to cope with the growing problems of private automobile-centered urban mobility systems. These problems include greenhouse-gas emissions, environmental pollution, oil dependency, land consumption for road and parking space, and urban congestion. The benefits of shared mobility originate from reducing the total number of automobiles at the expense of convenience of the private automobiles.

However, shared mobility also faces various challenges based on the service model. One-way car rental services and mass transits encounter the first and last mile problem. Mobility-on-Demand (MoD) services were proposed to offer transportation between transit stations and homes/workplaces [7]. One successful example of MoD, the OV-Fiets system in the Netherlands, was implemented to offer commuters access to bicycles for traveling from the train station to their destinations [3]. MoD services inherently have a vehicle asymmetry problem between supply and demand [20]. The use of autonomous driving technology has been used to solve the first and last mile problem [8] and well as vehicle rebalancing [14].

The convergence of the Internet, wireless communications, geographic information system, location technologies, and mobile devices has contributed significantly to the recent success of demand-responsive transportation services including ride-sharing services and optimal taxi assignment [5], [16]. In a similar context, autonomous valet parking was proposed and implemented using the aforementioned technologies in [21].

From the perspective of mass transit, driverless trains have been adopted for their benefits to safety and capacity on demand, allowing automated trains to adapt to passenger demand immediately without the need for additional operational staff and thus eliminating the risk of human error [11]. CityMobil2 is a driverless bus designed to take passengers to the nearest mass transit station in cities, where demand is low or pick-up points far apart [17]. Pendleton *et al.* demonstrated passenger transport using autonomous golf cars at a course in a public park [18]. Brownell mathematically modelled and analyzed transportation demand in shared autonomous taxi networks based on traffic data in New Jersey, United States in [22].

In Table I, all the shared mobility services described are summarized using comparative information, where *FM* and *LM* are checked if the first mile problem and last mile problem, respectively, are solved by the service. *RB* is ckecked if there are no vehicle rebalancing issues. *PP* is checked if pick-up and drop-off points can be separated from the transit station, indicating point-to-point mobility. Lastly, *AT* is checked if automated vehicle technology is involved.

There have also been many research efforts to enhance campus mobility. Each campus has unique challenges in terms of geographic location, campus layout, and mobility demand and patterns. One popular suggestion for sustainable

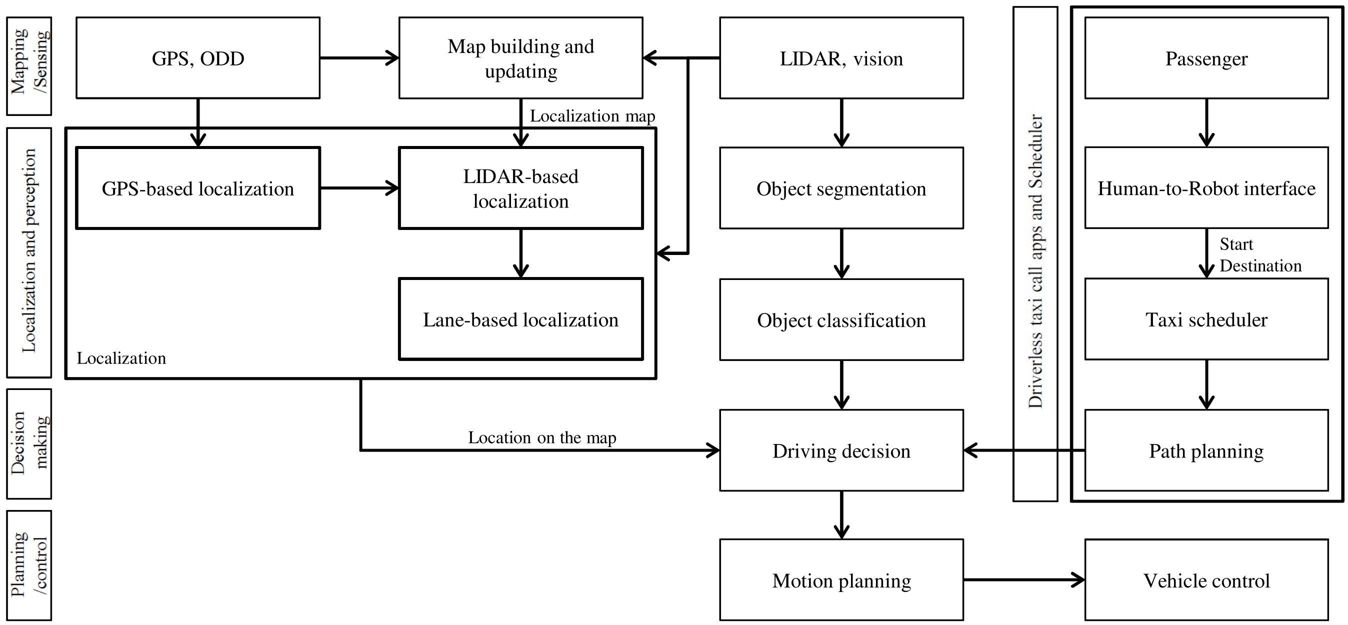


Fig. 1. Overall ACMS architecture. The top-right section relates to customer requests and taxi assignments. The remaining sections relate to the operation of a driverless taxi: mapping, localization, perception, decision making, planning and control.

campus mobility is to develop and implement a bicycle and pedestrian friendly policy [12]. However, this may not be widely adaptable for rural campuses. An urban campus tends toward vertical connectivity, whereas a rural campus tends toward horizontal connectivity and is more automobile dependent [13]. Rural campuses often include sloped areas. For example, the altitude of SNU campus ranges between 63 m and 161 m. Weather conditions such as temperature, humidity, and precipitation are also important factors. These factors have inhibited wide-spread adoption of bicycle and pedestrian friendly policies. Although campus shuttle bus services are typically provided, they encounter the first and last mile problem. Conventional taxis are also a suboptimal alternative for campus mobility.

In this context, there are several benefits of ACMS compared to a taxi with a driver. Firstly, driverless taxis are expected to be owned by mobility service providers because of 1) additional high cost of autonomous driving functions and 2) integration of service management systems. In the mobility service model, insurance premium can drop by eliminating risk factors affected by individual drivers. Secondly, driverless taxis do not get tired, which increases vehicle utilization and decreases road accidents caused by driver fatigue. Thirdly, driverless taxis are expected to have less running cost by integrating them into electric vehicles. Fully-autonomous driving works well with electric cars due to their born-naturally X-by-wired and better controllability, even though the both are developed in parallel. Running cost of an electric car is less than an equivalent combustion powered car. Lastly, autonomous driving works well with collaborative driving, which have been addressed in [23]–[25]. For these reasons, this paper proposes a driverless taxi to provide point-to-point campus mobility. Note that there are as many new things to consider as the benefits, which are addressed in Sec. V.C and Sec. V.D.

We describe the details of our autonomous campus mobility service in the following sections

III. Autonomous Campus Mobility Service

The ACMS is a demand-responsive campus mobility service using autonomous driving vehicles. The overall service scenario can be briefly described as follows.

A user begins by opening the ACMS mobile app on their mobile device. Their current location is automatically obtained by the GPS (Global Positioning System) in the mobile device. The user destination is then selected using the on-screen map. The mobility service request enters the queue on the ACMS central server. The central server then receives a request from the queue and assigns the request to a driverless taxi. This process is shown in the top-left corner of Fig. 1.

The remaining operation is to perform autonomous driving to the pick-up point and destination, which corresponds to the remainder of Fig. 1. The assigned taxi autonomously drives to the pickup location specified in the request. After arriving at the pickup location, the user gets in the taxi and pushes the start button on the app. The taxi then autonomously drives to the destination. After arriving at the destination, the user gets out of the taxi, and the taxi drives to the next user.

As for the driverless taxi, we have built an autonomous driving vehicle equipped with a 3D Lidar (Velodyne HDL-64E), a GPS+INS vehicle positioning system (OXTS RT3002), and a vehicle-mounted vision camera, as shown in Fig. 2. The key features of the Lidar are a 120 m range, 64-layer, 360 degrees horizontal FOV (Field-Of-View), 26.8 degrees vertical FOV, and a 0.08 degrees angular resolution (azimuth), which is configured to measure 1.3 million points per second at a 10 Hz rotation rate.

The Lidar is mounted on top of the taxi, and is used for map building, map updating, localization, and vehicle/pedestrian detection. The mono camera is installed behind the windshield

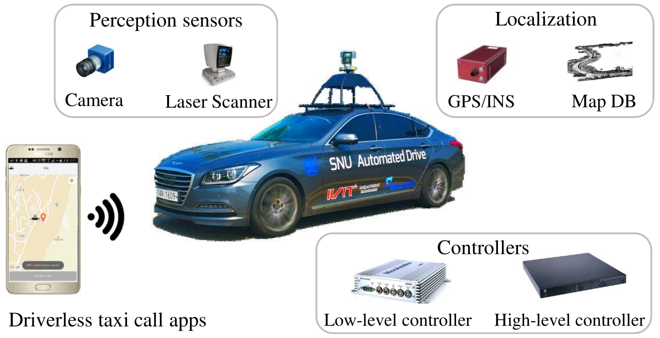


Fig. 2. System configuration with perception sensors, localization modules, controllers, and smart devices.

and detects roads marking such as lanes, stop lines and speed bumps. The camera can classify the types of objects and interpret traffic signs and traffic lights. The OXTS RT3002 is a unified GPS+INS system that provides 6D pose and is used to assist in map building and localization.

Acceleration, deceleration, and steering are fully controlled by the Control Area Network (CAN). Two computers with 3.40 GHz i7-4770 CPUs are installed in the trunk and used to operate our software modules. The computers use environmental sensor data and in-vehicle sensor measurements to run algorithms for modules that will be explained in detail in the following section. The algorithms recognize obstacles and carry out path planning, allowing the vehicle to drive efficiently and avoid collisions.

The path planning messages are sent to an Autobox, which performs vehicle control using the actuators. Gigabit Ethernet cables connect the environment sensor–computer and computer–computer, and CAN buses are used to connect the computer–Autobox, Autobox–in-vehicle sensor, and computer–in-vehicle sensor. A user request is trasmitted to the ACMS through 3G HSDPA (High-Speed Downlink Packet Access), and 4G LTE (Long Term Evolution).

IV. System Architecture

In this section, we provide a technical description of the driverless taxi and ACMS. Fig. 1 presents the overall system architecture.

A. Map Building

Our autonomous driving system relies primarily on a precise 3D digital map for localization and planning. As a first step towards creating such a map, we designed and developed a 3D mapping and map updating system using a 3D occupancy grid map with a 3D laser scanner. Fig. 3 presents the overall process for map building and updating. The method introduced here can generate a precise 3D map at centimeter-level resolution without GPS availability, even in a tunnel, comparable to previous methods, e.g., [26], [27].

In Fig. 3, mapping results have spurious trails caused by moving objects. In order to detect and track moving obstacles, a ray-casting approach is used [28]. As a result of moving object removal, a free-space is extracted. To this end,

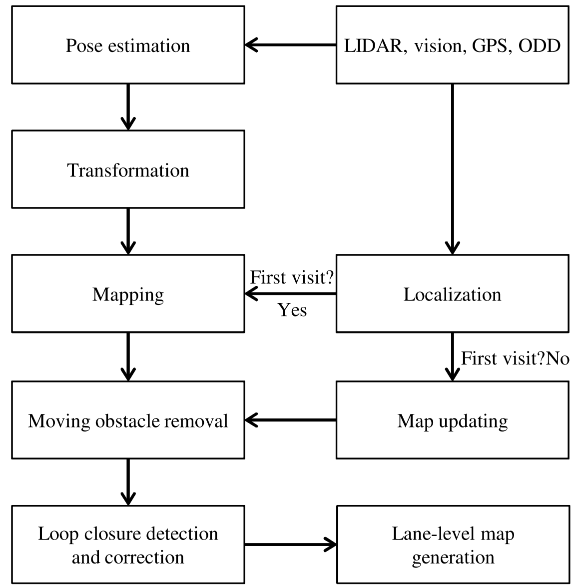


Fig. 3. Process flow of map building and updating.

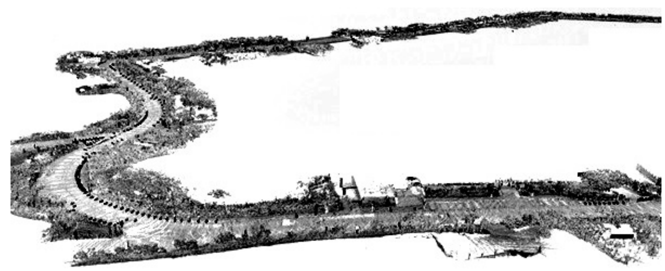


Fig. 4. Moving object removal results using the ray-casting method.

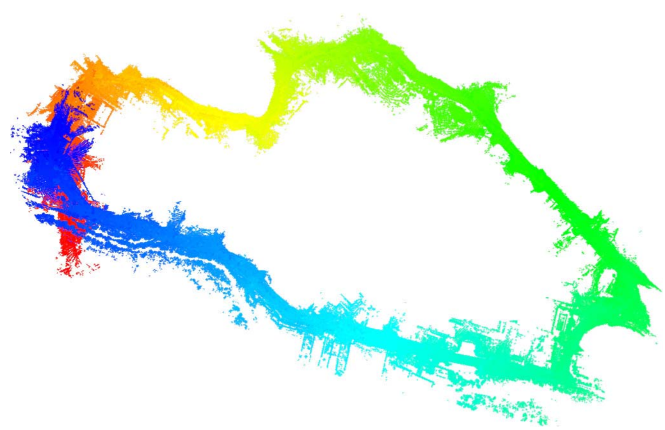


Fig. 5. Sequential scan matching results without loop closure correction.

the 3D occupancy grid map includes only static information. Fig. 4 displays the mapping results of moving object removal. However, semi-static objects, such as parked vehicles and stationary pedestrians, are still regarded as static objects. In order to remove such semi-static objects, a mechanism for map updating is necessary. A map is updated if the mapping vehicle revisits a previously visited area.

Fig. 5 displays the sequential scan matching result for a 4.5 km SNU campus road. In Fig. 5, the colors represent the

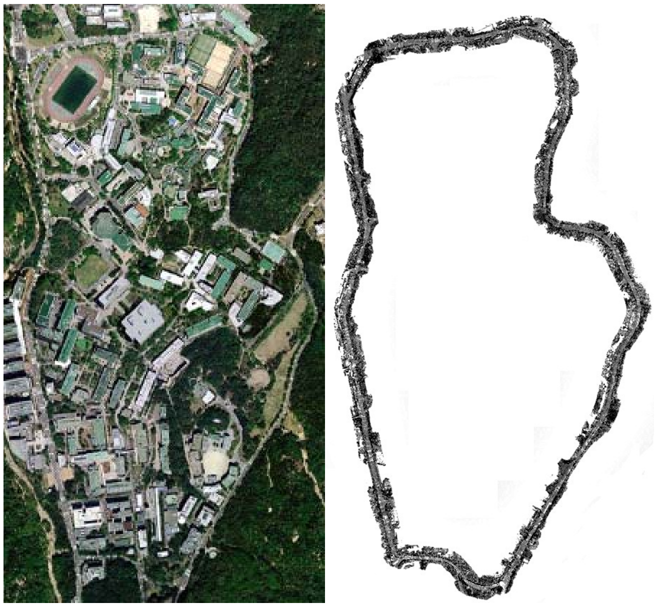


Fig. 6. Loop closed map of the Seoul National University ring road.

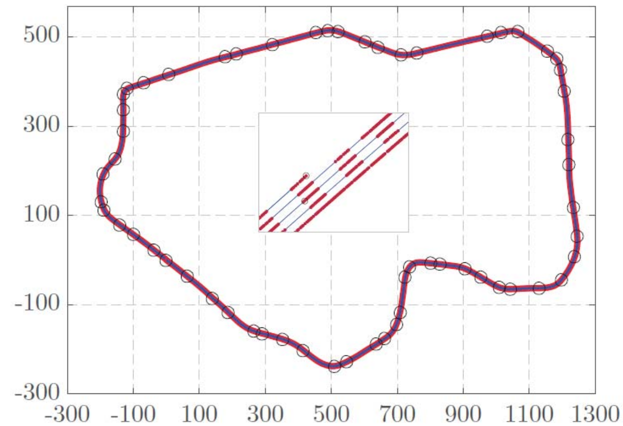


Fig. 7. Lane-level map of the Seoul National University ring road generated from the loop closed map in Fig. 6. Figure courtesy of authors in [30].

ordering of scanned frames. In principle, the blue area and the red area should be closely matched. Due to the accumulation of sensor noise, matching errors, and chained multiplications of transformation matrices, the loop is not closed. In order to solve this problem, we developed a new global relaxation scheme.

First, we use GPS information to detect a loop. We detect a loop from the ith frame to the jth frame if the following conditions are satisfied:

- The number of available GPS satellites is sufficient during data collection from the ith frame to the jth frame.
- The GPS coordinates from the ith frame to the jth frame are close enough to calculate an overlapping area.
- A sufficient distance exists between the ith and jth frames.

If the above conditions are satisfied, a relative pose between the ith and jth frames can be calculated through scan matching. Loop closure is not satisfied, if the relative pose is nonzero. In order to achieve loop closure, the loop closing error can be equally divided across all frames as proposed in [29].

However, this method does not perform well for large loops such as the campus road introduced in this paper. We instead assume that the rigid motion derived from scan matching includes a small regular error due to the inherent distortion in scanner data. Under this assumption, there exists a correction matrix that can correct the scan data distortion of a loop. The correction matrix can be found by solving the global relaxation for loop closure using a Monte Carlo method. Technical details of this is out of scope in this paper. Fig. 6 shows the loop closed map from Fig. 5.

In order to make a map scalable and readily accessible, a lane-level road map is generated from the 3D map to meet the requirements of autonomous driving, such as low processing-time, high storage-efficiency and usability. For example, three billion points are collected over the 4.5 km long road in our setting. In order to build a lane-level map from a 3D map built by using the aforementioned method, the points corresponding to lane markings are extracted and clustered with the points that belong to the same lane. The set of the lane marking points corresponding to each lane is then modeled as a spline curve, which is composed of cubic piecewise polynomial curves that are parameterized by arc length to improve storage efficiency for the map. Through this lane-level map generation process, the three billion points are reduced to 30 000 points. The lane-level map is presented in Fig. 7, which is generated from the loop closed map in Fig. 6. One can find the details of our lane-level map building process in [30].

B. Localization

Unlike in GPS-navigation systems for driver assistance, it was shown that GPS alone is insufficient to meet localization requirements for autonomous driving [31]. For example, autonomous valet parking [21], or autonomous driving in a tunnel or a forest of skyscrapers, are all nearly impossible with pure GPS localization. Precise localization at a centimeter-level resolution is a key requirement for fully autonomous driving.

While satisfying the centimeter-level requirement, various road characteristics should also be considered for localization. The characteristics of a road vary according to the function of a road, e.g., highway, well-structured road, urban/rural road with complex environments. In order to enable localization for various roads, ACMS uses 1) 3D map-based localization for complex urban/rural environments, 2) localization with a lane-level map, and 3) lane-level localization without a map for well-structured roads.

1) Localization With a 3-D Map: For localization, ACMS uses a particle filter [32] on the 3D map generated in the previous subsection. The 3D map is projected onto a 2D x-y plane for localization proposed in [26]. Based on the 3D-to-2D mapping system, we must account for semi-static objects such as parked cars and stationary pedestrians. At SNU, street parking is permitted due to a shortage of parking spaces, which results in the presence of semi-static objects during map building and localization.

This makes it difficult to perform localization through curb detection [33] and map matching, which have been commonly used for vehicle localization. Even though a probe vehicle is roaming around the campus to detect any changes in the map, it is impossible to reflect every change to the map in real-time due to the unpredictable mobility patterns of the semi-static objects. We develop a localization method inspired by a temporary map to cope with this problem [34].

2) Localization With a Lane-Level Map: Our vehicle performs lane-level localization using the lane-level map addressed in the previous subsection. The strength of a lane-level localization with a precise map is the ability to perform localization with affordable off-the-shelf sensors such as in-vehicle cameras and a low-cost GPS. In particular, the geometry information of lanes is useful for increasing lateral localization accuracy. Lane-level map-based localization is performed by integrating the lane-level road map with various sensors, including a camera, on-board vehicle motion sensors, and a low-cost GPS using a particle filter-based algorithm.

The core of the localization algorithm is matching between lane marking information in the camera image and the road map. In our algorithm, the road map data near the ego-vehicle is transformed into the image plane with respect to certain candidates for true vehicle position, and the matching scores between the transformed road map data and the lane markings in the image are calculated. Then, the optimal estimate of the vehicle position is calculated using a particle filter framework. One can find the technical details in Sec. VI.D of [30].

3) Localization Without a Map in Well-Structured Road: Even without a prior map, our system can perform lane-level localization in an alternative manner. Two algorithms are required to achieve this goal: multi-lane detection [35], [36] and lane-identification [37]. The multi-lane detection algorithm provides lane model information, including the ego-lane's offset C_0, heading angle C_1, and curvature C_2 for direct driving, whereas the lane-identification algorithm provides lateral localization information about the lane in which the ego-vehicle is driving. Both algorithms rely largely on visual sensors and yield results directly utilizing consecutive frames.

These algorithms are performed sequentially because they share initial lane detection steps. First, the multi-lane detection algorithm adapts Conditional Random Fields (CRF) to associate each supermarking with its corresponding lane-label, where the supermarking is defined as a cluster of neighboring lane features in [35]. After association, all markings in the same lane-label are fitted using RANSAC (RANdom SAmple Consensus) to derive a second order polynomial and tracked using a particle filter, where a lane is modelled as a second order polynomial [36].

Subsequently, the lane-identification algorithm collects additional lane characteristics from the multi-lane detection results such as lane shapes S, colors C and lane-existence configuration F. All these observations $\Im = \{S, C, F\}$ are integrated in a Bayesian Network model [37] to determine the ego-lane's index. One prior, the ego-lane's index $E = \{Opposite, 1st, 2nd, Shoulder\ lane\}$, is classified based on the model Eq. (1) and pre-trained conditional probabilities.

$$P(E, \Im|\Theta) = P(E|\Theta)\prod_{i=1}^{N_C} P(c_i|E, \Theta)\prod_{i=1}^{N_S} P(s_i|E, \Theta)P(F|E, \Theta), \quad (1)$$

where Θ is the set of all parameters in the model. N_C and N_S are the cardinality of the sets C and S, respectively. When these algorithms are executed, global lateral offset from the centerline can be calculated using the equation $L_O = w(E-1)+C_0$, when w is assumed to be an average lane width. One can find the technical details in [37].

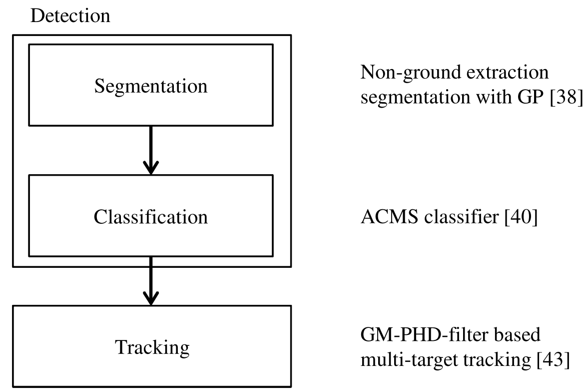


Fig. 8. Perception process.

C. Perception

A campus road may contain many walking pedestrians and driving vehicles. Thus, detection and tracking of moving objects is indispensable for autonomous campus mobility services in terms of safe driving and decision making. Fig. 8 presents the overall flow of the perception process used for this purpose, which is comprised of segmentation, classification, and tracking.

1) Segmentation: Our segmentation algorithm plays a key role in providing a basis for the other phases of perception. Typically, segmentation performance is degraded by over-segmentation that separates one object into multiple segments, which in turn lowers the performances of the other perception algorithms operating based on the segmentation result. We developed a method to handle the over-segmentation problem based on Gaussian Process (GP) regression and proved that the method reduced over-segmentation significantly. Based on the decreased over-segmentation, our algorithm is able to achieve high segmentation accuracy and performance improvements in object tracking of 10–20% on average [38].

Additionally, our algorithm achieves real-time processing that is very competitive when compared to the previous fastest segmentation algorithm presented in [39], requiring approximately 40.1 ms for 130 000 data points on average. In our algorithm, this high speed is obtained largely from skipping unnecessary computation steps, such as ground extraction. In the previous algorithm, processing speed was obtained at the expense of performance degradation due to the additional

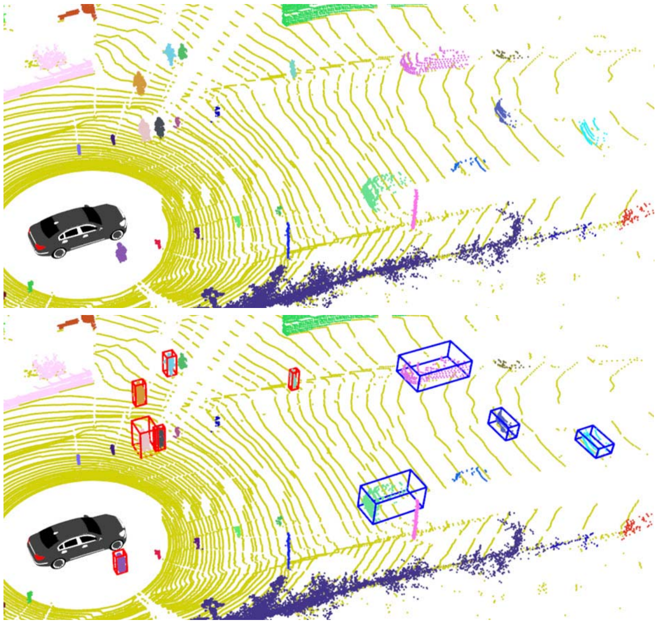


Fig. 9. Pedestrian and vehicle detection on a campus road.

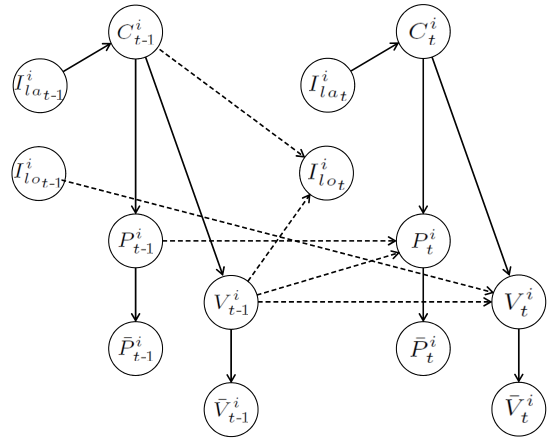


Fig. 10. Graphical representation of the intention inference model. Dashed lines correspond to time dependencies.

techniques employed for speed-up, such as dimensionality reduction of 3D points. Our segmentation algorithm accomplishes both real-time processing and high accuracy at the same time.

2) Classification: The objects present in the the segmentation results are classified as static objects, semi-static objects, or moving objects. Candidate objects are extracted based on their size and their ratio of height and width [40]. The filtered candidates are classified based on a combination of ten types of features and a classifier designed using a Support Vector Machine (SVM) with a Radial Basis Function (RBF) kernel. Some example features are the ratio of the height and width of an object, the 3D covariance matrix of the cluster, the normalized moment of inertia tensor, the distribution of the reflection intensity, etc. A few of these features were proposed by Navarro-Serment *et al.* [41], and Kidono *et al.* [42]. One can find the technical details in [40].

3) Tracking: The classified objects are tracked by our multi-target tracking algorithm. It is essentially a modified version of the Gaussian Mixture Probability Hypothesis Density (GM-PHD) filter, which is a well-known Multi-Target Tracking (MTT) algorithm. Our algorithm remedies some of the inherent problems in the original algorithm. It was observed that the original GM-PHD filter produces over/underestimation when false positives or negatives occur frequently. For example, in the event that an object is not continuously detected over a few frames, the weight for that object will become extremely small because the algorithm decides that the object is gone. In this case, the object is difficult to track even if it is perceived again. In order to solve this type of problem, we introduce the notion of a confirmation score in our algorithm to designate the statuses of tracked objects as unconfirmed, confirmed, or missing. The algorithm adjusts the weight of tracked objects properly for each status. For additional details about our algorithm, refer to our publication [43]. Fig. 9 is the result of pedestrian and vehicle detection and tracking results. No cyclist were observed due to the steeply sloped road of the campus.

D. Decision Making

Driving involves continuous decision making until the destination is reached. One characteristic of campus roads that differ from other roads is the frequent occurrence of unsignaled intersections with heavy traffic, and two-lane roads with no median barrier. Therefore, our decision making system for driving through the unsignaled intersections and overtaking a static object by violating the centerline is essential for automated driving on campus roads. This subsection introduces how to pass through intersection autonomously and overtake a static object on a two-lane road without the support of wireless communications.

The automated vehicle should act in a manner robust to uncertainty. In order to overcome uncertainties, automated vehicles generally make a decision based on a probabilistic approach, such as a Bayesian Network (BN) or Partially Observable Markov Decision Process (POMDP) [44], and semantic level recognition results, such as driving intention [23]. For decision making at unsignaled intersections, the process is subdivided into the following steps.

First, the decision making system infers the driving intentions of other traffic participants. This is accomplished using a geometric model from the pre-defined precision map to enable a comprehensive intention-aware decision making system. The evolution of traffic situations and driving intentions are then modeled as a stochastic process. This process is represented by a Dynamic Bayesian Network (DBN), where the layout of the intersection, the traffic rules, and the dependencies between vehicles are explicitly accounted for [45]. Three categories of variables are defined to model intention awareness by using DBN: physical variables, behavioral variables, and intention variables, as shown in Table II. Fig. 10 depicts a graphical representation of our intention inference model using these variables.

Given the intentions inferred by the DBN, the intention-aware decision making problem is modeled as a POMDP [46],

TABLE II
PARAMETER DEFINITION FOR INTENTION AWARENESS

P_t^i	True pose of a vehicle i
$\bar{P}_t^i$	Observed pose of a vehicle i.
V_t^i	True velocity of a vehicle i
$\bar{V}_t^i$	Observed velocity of a vehicle i
C_t^i	Maneuver of a vehicle i's driver
$I_{la_t}^i$	Lateral driving intention of a vehicle i
$I_{lo_t}^i$	Longitudinal driving intention of a vehicle i

which is one of the most widely used models for decision making problems in uncertain environments. In this work, the vehicle receives a high penalty when a collision is predicted, a high reward when it is predicted to arrive at its goal, and low penalty for each time step. POMDP in this work makes a decision largely based on which longitudinal speed control action will result in the highest reward. With this reward model, the vehicle can pass through an unsignaled intersection as fast as possible while avoiding collision.

The automated vehicle in this work does not always overtake a stopped vehicle by violating the centerline. If the stopped vehicle is regarded as a permanently stopped vehicle, and there is no on-coming traffic on the opposite direction, the automated vehicle overtakes the stopped vehicle. A probabilistic BN model is used for inferring whether or not the detected vehicle is a permanently stopped vehicle. BN is one of the probabilistic graphical models that represents the mathematical principles of graph and probability theory in [47]. Nodes represent that each random variable from the domain of interest and edges represents the direction of influence of a parent node through an intuitive graphical representation in the form of a directed acyclic graph.

The parameters represents the probabilities of the nodes: prior probability for a node with no parents and a Conditional Probability Table (CPT) for a node with parents. We obtain results for size, stoppage time, and lateral distance from center of the driving lane of the vehicle, and as well as the number of stopped vehicles around the ego-vehicle, from the perception system. These four features provide information to observation nodes and the BN integrates all the data in a structured manner. The goal of a BN is to infer the states of unobservable variables, such as whether or not a detected vehicle has stopped permanently, given the evidence provided by the four observable variables mentioned above.

Once a detected vehicle is assumed by the BN to be a permanently stopped vehicle, and there is no on-coming traffic on the opposite direction, the ego-vehicle generates an overtaking path that will avoid collision with the stopped vehicle and violate the center line as little as possible. One can find the technical details in [45].

E. Planning and Control

The ACMS planning system consists of path planning and motion planning subsystems. The path planner generates an optimized trajectory for the assigned taxi when the users current location and destination are provided. The trajectory

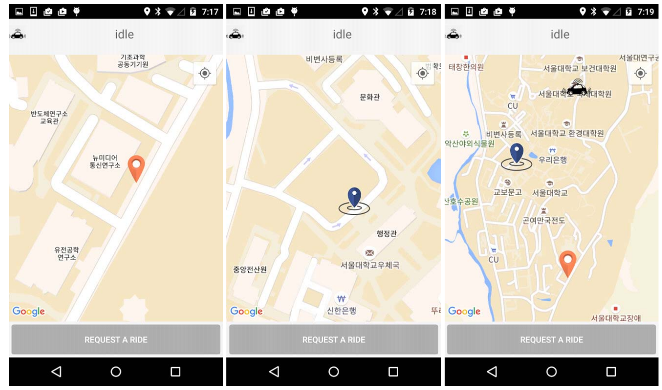


Fig. 11. User interface. The map pin automatically sets a start point through GPS (left-most), a user sets the destination on the map (middle), and the user can check the status of the driverless taxi. (right-most).

is optimized in terms of length, semantic map information and the schedule received from the taxi scheduler. The motion planner generates a local path, which is determined by collision avoidance decisions within the drivable space. We devise and use a RRT* [48] based motion planner for the purposes of our service.

When the optimized trajectory is determined by the planning system, a controller is needed to accurately track the trajectory and perform safe driving. At each time-step, the controller performs two controls: desired speed and steering angle. Desired speed is determined by the driving contexts. In our work, the driving context is the predicted trajectory of moving objects, comfort factor and semantic information, such as traffic regulations. For example, the maximum velocity of our taxi is 30 km/h due to the speed limit on the SNU campus. A Proportional-Integral-Derivative (PID) controller [49] is used for velocity tracking, which minimizes the error between the desired and current velocity. The desired steering angle is determined by the current status of the taxi and the desired trajectory. A pure pursuit method, which considers geometric relationships between the vehicle and the path, is used in our work.

F. App: Customer Interface

As shown in Fig. 11, the user-interface window is separated into three section: top, middle, and bottom, which represent taxi status, a geographical map, and user command buttons, respectively. The number of user command buttons at the bottom changes depending on the state of the user, which can be *Idle*, *Awaiting*, and *Onboard*.

When the user state is *Idle*, indicated that the user is ready to call a taxi, the only button visible at the bottom is *Request a ride*. After pressing the button, a driverless taxi, if available, comes to the current GPS location of the user. Alternatively, a user can set a custom pick-up location, and choose their destination, by long-pressing the map in the middle. The next state after pressing the request button is to wait for the taxi to arrive. In this state, the button changes from *Request a ride* to *Cancel*, which allows a user to cancel the request.

Changing the destination is also possible by long-pressing the map in the middle. Once the taxi arrives at the pick-up

Fig. 12. Autonomous campus mobility service.

point, the *Cancel* button is split into two buttons: *Onboard* and *Cancel*. After pressing *Onboard*, the taxi considers that the user is now on board and ready to move. In this state, the *Ready to Go*, *Stop*, and *Off* buttons are displayed. *Ready to Go* commands the driveless taxi to start autonomous driving toward the destination. The user commands the driverless taxi to pull over by pressing the *Stop* button. When the taxi arrives at the destination as requested, or stops along the road due to a press of the *Stop* button, the user can get off the taxi. The service is then ended by pressing *Off*.

V. Test Results

The first public demonstration was held at SNU on November 4, 2015 [51]. Fig. 12 displays the ACMS in operation at the time. An SNU campus 3D maps was built prior to the demonstration as presented in Figs. 6 and 7. The altitude difference between the lowest and the highest point is 98 m, which is the main reason why very few bicyclists are seen on SNU campus. Since the first demonstration, a regular service has run on a weekly basis to collect user feedback and improve user satisfaction. While we were testing the ACMS, a safety driver sat behind the wheel to take over in any potential emergency situations. There has been no human safety driver intervention during the operation to this point. The ACMS has been tested in the real traffic, which means no need to design a scenario and build the test road according the scenario.

A. Overall Performance

The performance of perception and control in our autonomous driving system is evaluated in the following manner. In order to evaluate the perception performance, such as recognition of moving objects that affect the planning and control of driving, *precision* and *recall* are adopted as performance metrics. *Precision* is the fraction of successful retrievals over all retrievals, while *recall* is the fraction of retrievals over the number of really objects, also referred to as sensitivity. *Precision* and *recall* are calculated using the following equations:

$$Precision = \frac{True\ Positive}{True\ Positive + False\ Positive},$$
$$Recall = \frac{True\ Positive}{True\ Positive + False\ Negative}.$$

TABLE III

Performance Comparison of Moving Obstacles on Campus Road

Algorithm	ACMS [38]	Douillard [50]	Himmelsbach [39]
Precision	0.970	0.915	0.943
Recall	0.958	0.948	0.950
Avg. processing time (ms)	40.1	115.5	41.8
Std. deviation (ms)	8.4	10.6	5.2

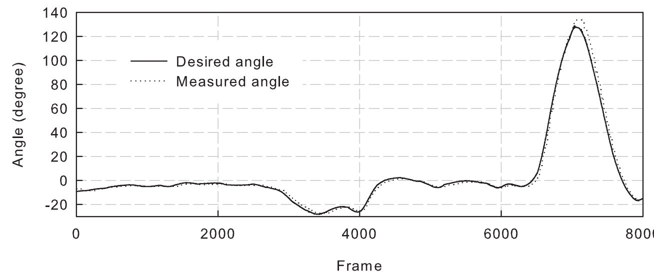


Fig. 13. Steering performance.

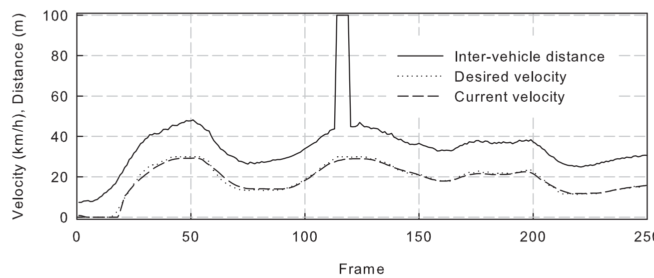
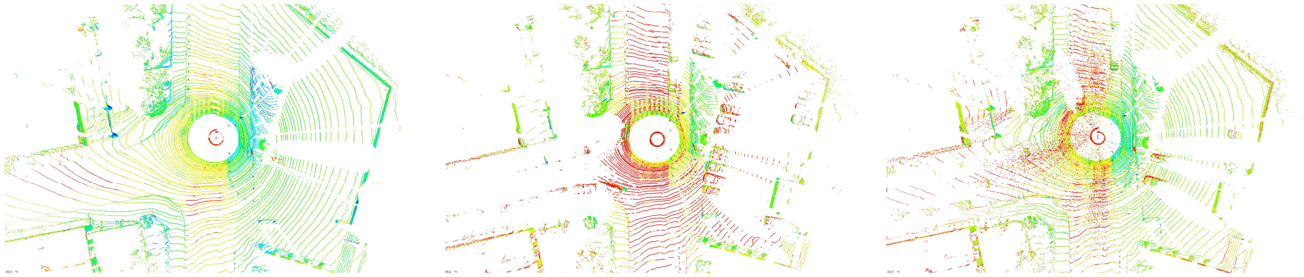


Fig. 14. Longitudinal control performance.

Table III displays the perception performance with a comparison to the results of Himmelsbach *et al.* [39] and Douillard *et al.* [50]. The perception algorithm adopted in the ACMS is enhanced by reducing over-segmentation, which allows it to achieve better results for both perception metrics. The processing time is the fastest in our campus environment. The details in perception algorithms and performance evaluation can be found in [38].

Fig. 13 displays how well our steering mechanism follows the desired steering angle generated by the planner in Sec. IV-E. The slight mismatch of the desired and measured angle is mainly caused by the steering torque of the steering wheel. It is a design problem to set an optimal steering torque. Strong steering torque of autonomous driving systems makes a safety driver difficult to take control back from autonomous driving system in any potential emergency situations.

Fig. 14 shows the performance of how well our speed control mechanism achieves the desired inter-vehicle distance that the planner generates. Note that the maximum velocity of our taxi is set to 30 km/h due to the speed limit on the SNU campus.



(a) (b) (c)

Fig. 15. Snapshots of laser scanning in (a) sunny/cloudy (b) rainy and (c) snowy weather.

B. Weather Impact on Autonomous Driving

For more than one year across all four seasons, through testing on the road, we have tested extensively in various weather conditions, including sunny, cloudy, rainy, and snowy. Weather conditions should be carefully considered for their potential impact on autonomous driving due to the performance disturbance of wireless communications and sensor measurements. This subsection presents the influence of weather conditions on autonomous driving in ACMS.

Fig. 15 displays snapshots of a single laser scanning frame in sunny/cloudy, rainy, and snowy weather, respectively, where the color of each point represents the strength of a reflected beam in the 0-255 range. Blue and red represent a higher and lower reflectance, respectively. The laser intensity represents the ratio of the strength of the reflected beam over that of the emitted beam, and is influenced mainly by the reflectance of the reflecting object. Reflectance varies with material characteristics as well as laser [52]. Certain road marking perception [30], feature extraction, and vehicle localization algorithms [53] use these intensity properties.

Fig. 15(a) was acquired on a dry and clean day with very little fine dust, where road markings such as crosswalks or traffic lanes tend to have a higher intensity than the road surface, which yields results similar to previous research [53]. However, the intensity is only average when compared to general road markings because of old road paint or low reflectance marking materials. Vegetation tends to have a relatively high intensity in clear weather.

Fig. 15(b) was gathered on a rainy day with 20 mm of precipitation. Most reflectance levels dropped because objects got all wet in the rain. In particular, the road surface and vegetation shows lower intensity than in the dry and clean weather conditions. Road markings, however, maintain the same intensity as on the sunny day. As a result, there is an obvious contrast between road marking intensity and road surface intensity. Despite the rainy conditions continuing during data acquisition, no rain drops were detected by the laser beam.

Fig. 15(c) was acquired in snowy weather with approximately 5 cm of total snowfall. Different from rainy weather, snowy and slush-mixed road conditions show uneven intensity. For this reason, the road surface and road markings are not easily separable by using intensity data. The intensity of vegetation is similar with the intensity in rainy weather, which is somewhat lower than in the clear weather. Atmospheric conditions in snowy weather are completely different from

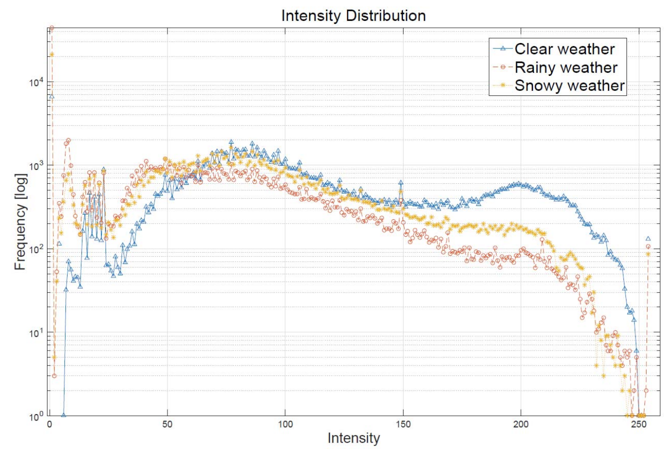


Fig. 16. Intensity distribution for each weather condition.

TABLE IV
AVERAGE INTENSITY FOR EACH WEATHER CONDITION

Quantity	Sunny or cloud	Rainy	Snowy
Average	108.1373	51.7547	74.9814
Standard deviation	59.6845	52.4826	54.2217

the other cases. As the snow particles are larger and fall more slowly than raindrops, they can be detected by the laser beam. These noise-like detected snowflakes and the non-uniform intensity of the road can potentially have a significant impact on autonomous driving, particularly in mapping and localization.

Fig. 16 shows the intensity distribution of each weather condition at the same location. The blue (triangle), red (circle) and yellow (plus) graphs represent the clear, rainy, and snowy weather, respectively. Even though the weather conditions are all different, the graph shapes are all similar. In particular, the transitional area are nearly the same in the intensity range from 20 to 150. Table IV shows the average intensity of each weather condition. The average intensity results correspond to the previous qualitative analysis. Although each data point comes from a completely different set of weather conditions, no significant difference is observed among the standard deviations.

We will now discuss the impact of weather conditions on mapping and localization. Figs. 17 and 18 show the mapping

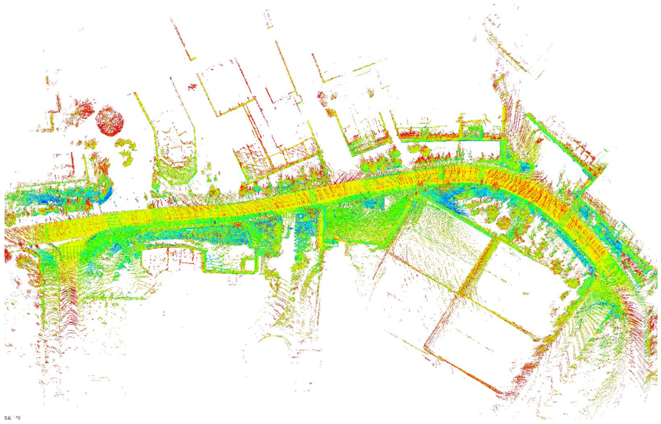


Fig. 17. Map building result from sunny or cloudy data.

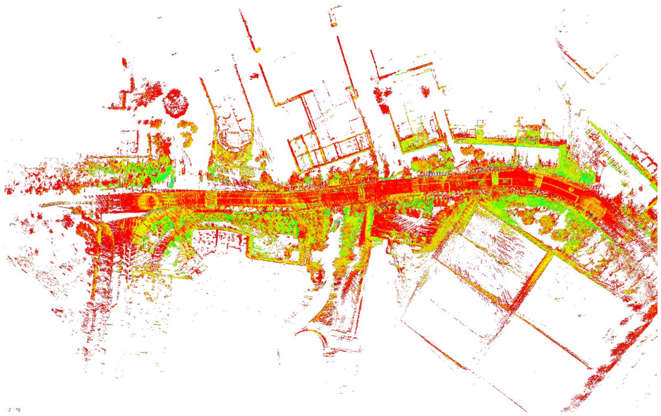


Fig. 18. Map building result from rainy data.

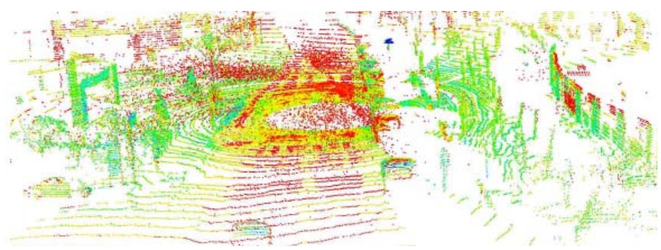


Fig. 19. Snapshot of laser scanning in snowy weather.

results generated from sunny/cloudy data and rainy data, respectively.

One important consideration for weather conditions is that they may change between mapping and localization. For example, mapping could be performed in cloudy weather, while localization could be performed in snowy weather using the cloudy map. In snowy weather, snow particles are detected by the laser scanner as shown in Fig. 19. Localization does not work well under these conditions. Prior to conducting localization, snowy data is post-processed to remove snow particles from the air as shown in Fig. 20.

C. Customer Feedback

In order to evaluate the customer satisfaction of the ACMS systems compared to conventional public transportation, such as campus shuttles, we conducted a customer satisfaction survey with ten customers who participated voluntarily in a riding test. Participants took a ride on the ACMS and campus shuttle on the same day in August during the summer.

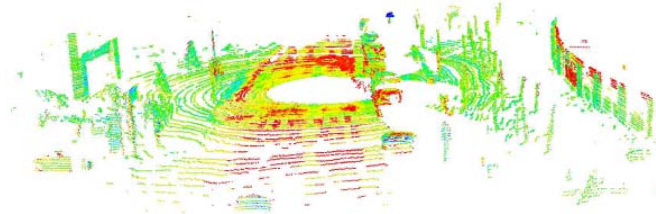


Fig. 20. Post-processing result of removing snow particles.

TABLE V
SERVICE QUALITY SURVEY RESULT

	ACMS	Shuttle
Reliability	4.5	2.8
Safety	4.1	1.8
Accessibility	0.8	4.2
Comfort	5	1.2
Convenience	3.4	4.1
Price	1.3	4.1

Note that this evaluation is intended to yield the tendencies and feedback of a focused group of customers, not mass customer satisfaction. Table V presents service quality survey results with the following metrics: *Reliability*, *Price*, *Accessibility*, *Comport*, *Safety*, and *Convenience*[1] Evaluation of these metrics was conducted using a Likert scale of 1-5.

The reliability of ACMS is nearly double that of the shuttle, which can be explained as follows. In the driverless vehicle, passengers can see 1) 360 degrees of traffic situations around the taxi, and 2) the taxi's reaction to any possible collision, which can affect reliability and safety. The safety of shuttle is 1.8 in Table V. It is guessed, because the driverless taxi is implemented on a premium sedan, which might make passengers feel that the shuttle is relatively less safe. Lower accessibility can be explained as the inconvenience caused by having only a single driverless taxi in operation. Currently, ACMS is provided for free of charge. Passengers were informed that this service would eventually charge fees similar to a normal taxi.

Additional feedback from customers can be summarized as follows. The speed limit is 30 km/h on the SNU campus, and ACMS has to stay strictly within the speed limit. Customers feel that an autonomous car is slower than human-operated cars that may violate the speed limit. This observation raises several issues including how to set the rules and parameters of autonomous driving in the real traffic. In this work, the driverless taxi always complies with the speed limit.

[1]Definition of Public Transport service quality attributes are as follows [54]. *Reliability*: How closely the actual service matches the route timetable. *Price*: The monetary cost of travel. *Accessibility*: The degree to which public transport is reasonably available to as many people as possible. *Comport*: How comfortable the journey is regarding access to seat, noise levels, driver handling, air conditioning. *Safety*: How safe from traffic accidents passengers feel during the journey as well as personal safety. *Convenience*: How simple the PT service is to use and how well it adds to one's ease of mobility.

Some customers mentioned that the reactions at intersections are too sensitive. This sensitivity originates from the long and wide perception capabilities of electronic sensors. Additionally, passengers desire smoother driving over speed bumps. A voice guide is essential, particularly when the service starts and ends. Without such notifications, the customers may not know why the ACMS stops, which could be due to arrival at the destination or simply waiting for a signal.

As for the service model, three major models have been proposed by customers: 1) ticket-based, 2) distance-based, and 3) semester-based service models. The ticket-based service model is similar to the ticket-based service model for mass-transit systems, such as a ticket for a one-way trip or a prepaid ticket. The distance-based service model is similar to a regular taxi. The semester-based service model corresponds to one day or one week of unlimited rides. Feedback regarding shared riding is positive as long as the riding cost is significantly reduced by sharing. We are continuously testing the system and collecting data to find a reasonable ACMS model.

D. Issues Serviced Out of Campus

In ACMS, the autonomous vehicle is programmed to comply with the traffic rules, which restricts the vehicle from taking any risk during driving. The long-ranged perception and 360 degree perception capabilities make the vehicle very sensitive, which causes the vehicle to stop more frequently than a normal human driver. It has been observed that such passive driving has caused traffic congestion during the test, particularly at unsignaled T-junctions with heavy traffic. Although our vehicle is equipped with a function for unsignaled intersection crossing through intention-awareness, it may not work well in cases of sudden heavy traffic. Sudden pedestrian crossing is still unpredictable as well, particularly from blind spots caused by parked cars. Additionally, ethical issues [55] must be clearly programmed into the vehicle based on the increased driving speed for a driverless taxi to operate outside of campus.

VI. Conclusion

In this paper, we introduced a campus mobility service using an autonomous vehicle. Our public trial of the driverless taxi on the road was very successful and required no human intervention during the service provided, and the ACMS raised public awareness of autonomous driving vehicles and their associated potential cultural changes. The service is currently in operation, and serves numerous requests from visitors, dignitaries, and campus users. Our vehicle has travelled over 10 000 km autonomously since the first public demonstration in November 2015. For the service proposed in this paper, our vehicle navigates the road primarily using 3D map based localization and perception, and partial vision-based lateral lane localization.

Through our user tests on the road, it has become clear that auditory and visual guides would be very helpful for users, because there is no driver present to explain various situation. On a similar note, a considerable number of users who have used the service multiple times wish to do more than quietly ride in our taxi. It is expected that more passengers will eventually be carried on each trip. We are also making efforts to expand to a fleet of vehicles. We believe that autonomous campus mobility services are important for the potential application of collaborative driving.

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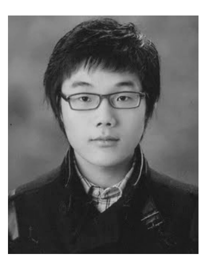

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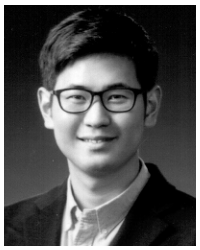

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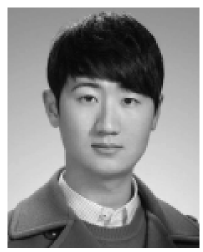

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